Three invariants

Quinn index · OpenAI Witt characters · Dabrowski L_4 tails

Preliminaries + short overview for the 2026 Hilbert–Smith signature arguments and their ENR boundary

Alan J. Coppola · AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models)

Roadmap

1. **Preliminaries** — signatures, Witt, L-theory, ENRs
2. **Geometry first** — what the algebra is really measuring
3. **Quinn / OpenAI / Dabrowski** — definitions
4. **Compare** — same Euclidean germ, two packagings

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

WHAT'S MISSING GEOMETRICALLY

The algebra is packaging. The geometry is a germ.

Quinn $I(X)$, OpenAI Witt characters, and Dabrowski L_4 tails look like different formalisms. Geometrically they all ask the same question:

*Can a tiny continuous $\mathbb{Z}_p$-motion of a **Euclidean (or ENR-dual) neighborhood** manufacture a "unit" of duality — degree 1 or signature 1 — and then split that unit into p^k equal covering pieces?*

If yes for arbitrarily large k, integrality breaks. Sponges fail earlier: they never produce the unit.

Two geometric moves (shared)

1. Chart compression $\to$ unit class. A small open subgroup of Homeo barely moves a ball. Haar-average / Newman rigidity $\Rightarrow$ a map that looks like the identity on a chart, then pinches to S^d or $S^n \times \mathbb{CP}^2$. That map carries *degree* 1 or *signature* 1 (the "unit").

2. Finite cyclic covers $\to$ p-splitting. Approximate $\mathbb{Z}_p$ by C_{p^k}. A free (or free-on-detector) C_{p^k}-action gives an orbit map of covering degree p^k. Transfer / characters split the unit into p^k equal pieces.

OpenAI writes (2) as Witt characters. Dabrowski writes (2) as $\tau = \mathbb{Q}[G/P]$ locally p copies of id. Same covering geometry.

WHAT'S MISSING GEOMETRICALLY

Why “characters” for both?

Not because Fourier analysis is the obstruction — because **free finite cyclic actions are covers**.

- Orbit projection $E \to E/C_{p^k}$ has degree p^k
- Transfer multiplies fundamental classes / signatures by p^k
- Characters of C_{p^k} are the Fourier language for the same isotypic splitting of equivariant forms
- Dabrowski’s permutation module $\mathbb{Q}[G_i/P_i]$ *is* the chain-level cover: locally p sheets

So OpenAI “equal parts $4/p^k$” and Dabrowski “eventually $p \mid \sigma$” are two bookkeeping systems for **covering degree**.

WHAT'S MISSING GEOMETRICALLY

Quinn $I(X)$: not "cobordism of $\mathbb{R}^n$"

Common misread: "high-dimensional cobordism classifies Euclidean space."

More accurate:

- An ENR $\mathbb{Z}$-homology manifold already has *algebraic* local Poincaré duality (like $\mathbb{R}^n$).
- **Resolution** asks for a cell-like map from a *genuine* topological manifold — geometric charts up to cell-like collapse.
- In dimensions ≥ 6, surgery / L-theory says that obstruction is a single integer $I(X)$ (local symmetric signature of the duality "corner").
- $I \equiv 1 \pmod 8$ is the Rokhlin-type shadow of 4-dimensional signature phenomena inside that local L-class.

$I(X)$ measures: "does local duality come from manifold geometry, or only from homology?" — not global cobordism of $\mathbb{R}^n$.

WHAT'S MISSING GEOMETRICALLY

Picture: corner vs chart

Manifold / resolvable ENR ($I = 1$): every point has neighborhoods cell-like equivalent to balls. Local duality is realized by actual geometry.

BFMW non-resolvable ($I \neq 1$): local homology still looks like $\mathbb{R}^n$, but the Poincaré corner carries an exotic local signature $I(X)$. No cell-like manifold resolution.

Stabilization: if $I(X) = 1$, then $X \times \mathbb{R}^2$ *is* a topological manifold — the remaining geometric obstruction is killed by two extra Euclidean directions.

How Quinn feeds the 2026 machines

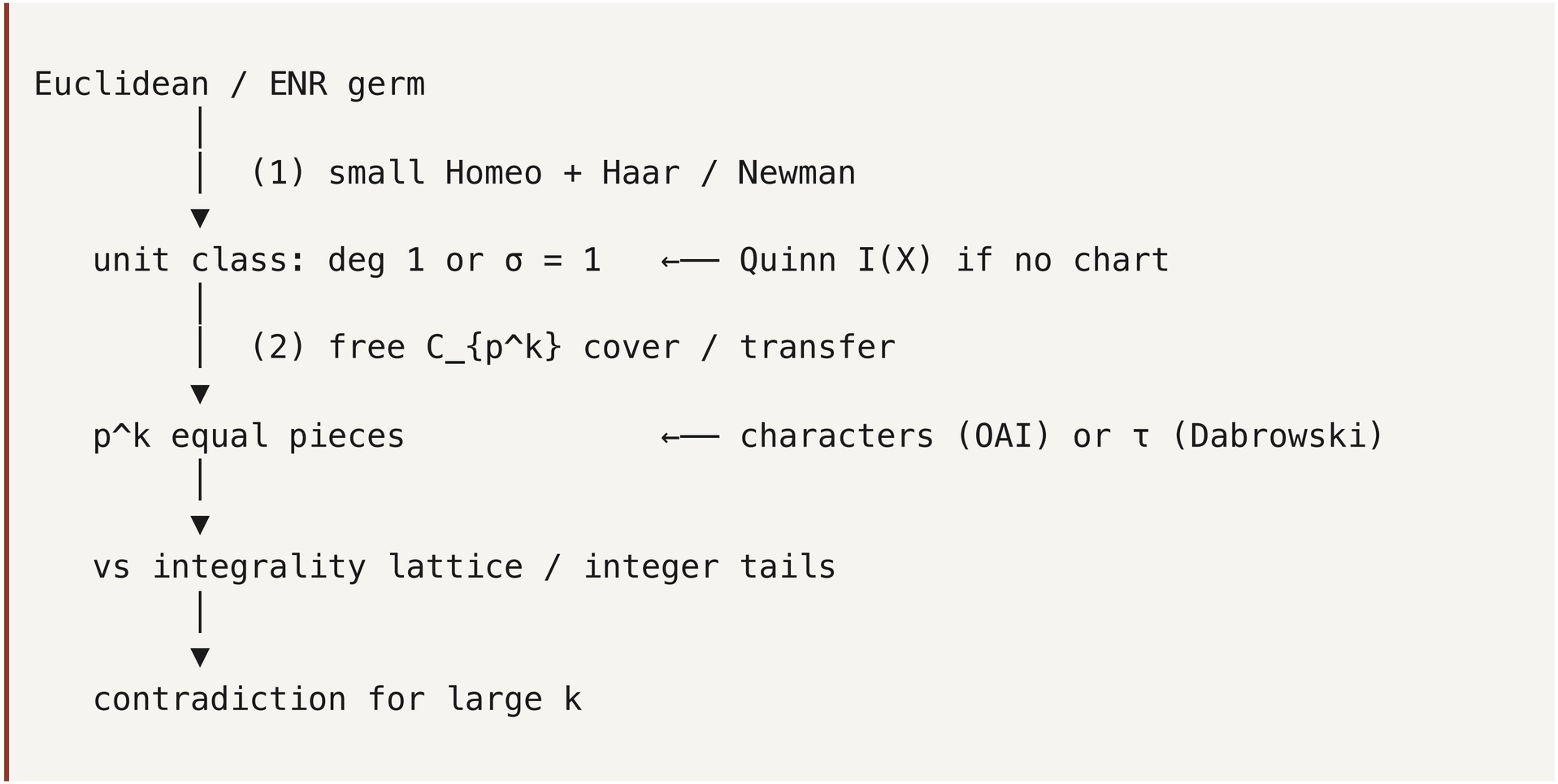
On a true manifold, the unit in move (1) comes from a Euclidean ball.

On a nowhere-Euclidean ENR, there is no ball — but there is still local duality. The 2026 ENR rephrases treat the detector's local signature as (a multiple of) $I(X)$:

- OpenAI equal parts: $4I(X)/p^k$ vs lattice L_d
- Dabrowski: local signature / tail linked to $I(X)$; single-p cyclotomic fails when $p \mid I(X)$

Geometrically: *Quinn replaces the missing chart's unit by the local L-signature of the homology-manifold germ.*

One diagram



Sponges / solenoids / RW compacta: fail at the top arrow (no unit-producing germ). Algebra never reached.

WHAT'S MISSING GEOMETRICALLY

So what was missing?

The missing geometric object is not a fancier invariant — it is the **unit-producing germ**:

- a neighborhood that can supply degree/signature 1 after tiny equivariant averaging, and
- compatible free finite cyclic covers that split that unit by covering degree p^k.

Witt groups, characters, L_4 tails, and Quinn $I(X)$ are three languages for bookkeeping that germ + those covers.

Without the germ, Hilbert–Smith packaging does not start. With the germ but only single-p arithmetic, exotic $I(X)$ divisible by p can slip through (Dabrowski gap).

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

PRELIMINARIES

What is a signature?

For an oriented $4k$-manifold M (closed), the **signature** $\sigma(M) \in \mathbb{Z}$ is the signature of the intersection form

$$H^{2k}(M;\mathbb{Z}) \times H^{2k}(M;\mathbb{Z}) \to \mathbb{Z}.$$

Number of positive eigenvalues minus number of negative eigenvalues (after tensoring with $\mathbb{R}$).

Classic: $\sigma(\mathbb{CP}^2) = 1$, $\sigma(S^4) = 0$, $\sigma(M \# N) = \sigma(M) + \sigma(N)$.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

PRELIMINARIES

Signature as a bordism / surgery invariant

- σ is a **cobordism invariant** of oriented $4k$-manifolds.
- In surgery theory, signatures (and their generalizations) obstruct whether a Poincaré complex is homotopy-equivalent to a manifold.
- For group actions / controlled topology, one needs *refined* signatures: Witt classes, L-theory classes, or "tails" of ordinary signatures.

The 2026 HS arguments manufacture such a refined signature from a Euclidean chart + small $\mathbb{Z}_p$ subgroup, then get a contradiction with p-divisibility.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

PRELIMINARIES

Witt group (classical)

The **Witt group** $W(R)$ of a ring R (e.g. $R = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$) classifies nondegenerate symmetric bilinear forms up to *metabolic* (hyperbolic) summands.

- $W(\mathbb{R}) \cong \mathbb{Z}$ via signature
- $W(\mathbb{Q})$ involves signatures at real and p-adic places
- A class can be "integral" relative to a lattice / denominator bound

Witt OpenAI works in a *sheafified / continuous* Witt group of forms on S^d, not just a single number σ.

PRELIMINARIES

Characters (for the OpenAI packaging)

A **character** of a finite abelian group G is a homomorphism $\chi : G \to S^1$ (or $\mathbb{C}^\times$).

- For $G = C_{p^k} = \mu_{p^k}$, characters are labelled by p^k-th roots of unity.
- Haar averaging + character projection splits an equivariant class into **isotypic / equal-size pieces**.
- OpenAI: p^k equal character pieces of a Witt/signature class on the target sphere.

Slogan: characters cut the signature "pie" into p^k equal slices.

L-theory in one slide

Ranicki L-groups $L_n(R)$ are surgery obstruction groups for a ring (with involution) R.

- $L_{4k}(\mathbb{Z})$ relates to ordinary signature
- For group rings / asymptotic categories, $L_4(A_G(T))$ carries *quadratic* Poincaré data for a detector space T
- "Ultimate" / Karoubi-localized L-theory forgets some control and yields ordinary integer signatures

Dabrowski uses L_4 of an asymptotic category, not a sheaf Witt group on S^d.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

PRELIMINARIES

Signature *tail*

A **signature tail** is a sequence $(\sigma_0, \sigma_1, \sigma_2, \ldots)$ of ordinary integer signatures, considered modulo sequences that are eventually zero:

$$\sigma_{\text{tail}} \in \mathbb{Z}^{\mathbb{N}} / \mathbb{Z}^{(\mathbb{N})}.$$

"Eventually p-divisible" means: for large i, $p \mid \sigma_i$.

Dabrowski's contradiction: realization gives the constant tail 1; divisibility says every tail is eventually in $p\mathbb{Z}$.

PRELIMINARIES

ENR homology manifolds

An **ENR** (Euclidean neighborhood retract) $X \subset \mathbb{R}^N$ is a retract of some open neighborhood of itself.

- **$\mathbb{Z}$-homology n-manifold:** $H_*(X, X \setminus \{x\}; \mathbb{Z}) \cong H_*(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ at every x
- Compact ENR $\Leftrightarrow$ finite-dimensional ANR (roughly)
- X_{man}: set of points with a Euclidean chart $\cong \mathbb{R}^n$
- BFMW: nowhere-Euclidean examples exist ($X_{\mathrm{man}} = \varnothing$)

PRELIMINARIES

Resolution of homology manifolds

A **resolution** of an ENR $\mathbb{Z}$-homology n-manifold X is a cell-like map $f : M \to X$ from a topological n-manifold M.

- Edwards–Quinn: for $n \geq 6$, resolvability is detected by a single integer.
- That integer is the **Quinn index** $I(X)$.
- $I(X) = 1 \Leftrightarrow X$ admits a resolution.

If $I(X) = 1$, then $X \times \mathbb{R}^2$ is a topological manifold — so Hilbert–Smith reduces to the manifold case.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

QUINN INDEX

Definition

For a compact connected ENR $\mathbb{Z}$-homology n-manifold X with $n \geq 6$:

$$I(X) \in \mathbb{Z}, \qquad I(X) \equiv 1 \pmod 8.$$

- Constructed as a **local symmetric signature** of the Poincaré duality "corner" of X
- $I(X) = 1 \Leftrightarrow X$ is resolvable (Edwards–Quinn)
- Product formula: $I(X \times Y) = I(X)\, I(Y)$
- BFMW construct examples with $I(X) \neq 1$ and $X_{\text{man}} = \varnothing$

Quinn Local L-signature of the duality corner — geometric resolution obstruction, not a global cobordism class of $\mathbb{R}^n$.

Examples of values

- Topological manifold / resolvable ENR: $I(X) = 1$
- Allowed congruence: $\ldots,\ -15,\ -7,\ 1,\ 9,\ 17,\ 25,\ \ldots$
- Non-resolvable BFMW examples: e.g. candidates with $I(X) = 9$ or 25
- If $I(X) \neq 1$, then $X \times Y$ is *never* a manifold for any generalized manifold Y with $I(Y) \equiv 1 \pmod 8$ (product formula)

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

Why it matters for $\mathbb{Z}_p$

On a nowhere-Euclidean BFMW space there is **no chart** — Newman reduction fails.

> Both 2026 arguments still hope to run on an ENR by treating the detector's local signature as (a multiple of) $I(X)$:
>
> Kill every $I(X) \neq 0 \Rightarrow$ no faithful $\mathbb{Z}_p$ on that ENR.

Resolvable case $I = 1$: reduce via $X \times \mathbb{R}^2$ to a manifold first.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

OPENAI PACKAGING

Witt lattice on S^d

OpenAI Chart $\to$ odd-stabilize into S^d $\to$ sheaf Witt / signature group $E_d(S^d)$.

Integrality: comparison lattice $L_d^{-1}\mathbb{Z}\,\overline{u}_d$ with fixed $L_d = 2^{e_d}$ (independent of the p-power k).

Generator: a preferred class $\overline{u}_d$ (orientation / signature unit on the target sphere).

Witt characters & equal parts

Haar-average a degree-one pinch through the orbit space. Character theory for C_{p^k} splits the class into p^k equal pieces:

$$\overline{z}_i(a) \;=\; \frac{4}{p^k}\,\overline{u}_d$$

On ENRs (rephrase): replace 4 by $4I(X)$.

For large k with $p^k > 4\,L_d\,|I(X)|$, this cannot lie in the comparison lattice unless $I(X) = 0$.

Slogan: p^k equal character slices vs a lattice with bounded denominators.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

DABROWSKI PACKAGING

L_4 signature tails

Dabrowski Same obstruction family, different packaging.

Object: ultimate quadratic $L_4\big(A_G(T)\big)$ for a free C_p-detector T.

Tail: ordinary integer signatures in $\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$.

Divisibility: every class has eventually p-divisible signatures (Karoubi / controlled Mayer–Vietoris; $\tau = \mathbb{Q}[G_i/P_i]$ locally p copies of id).

Realization: chart + Haar + $\mathbb{CP}^2$ germ $\Rightarrow$ a class with signature tail constantly 1.

Hence $1 \in p\mathbb{Z}$.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

DABROWSKI PACKAGING

Cyclotomic detector (single p)

Lemma 3.1: one index-p subgroup step uses $\Phi_{p^a}(1) = p$, so only **single-p** divisibility of the tail / local signature.

On ENRs: if the local signature is Quinn's $I(X)$ and $p \mid I(X)$ (e.g. $I = 9$, $p = 3$), then $I(X) \in p\mathbb{Z}$ *fails to contradict*.

Repair: k-step cyclotomic tower $\Rightarrow p^k \mid I(X)$; choose $p^k > |I(X)| \Rightarrow I(X) = 0$.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.

COMPARE

Side-by-side

	Quinn	**OpenAI Witt**	**Dabrowski**
Lives on	ENR $\mathbb{Z}$-hm	$E_d(S^d)$ / ENR pinch	$L_4(A_G(T))$
Value	$I(X) \equiv 1 \pmod 8$	class in 2-power lattice	integer signature *tail*
p-input	—	p^k equal character parts	τ / $\Phi_{p^a}(1) = p$
Contradiction	$I \neq 1$ non-resolvable	parts vs L_d	tail 1 vs $p \mid \sigma$
ENR gap	input	kills all $I \neq 0$	fails if $p \mid I$ (single p)

COMPARE

Takeaway

- **Geometry:** unit-producing germ (chart or Quinn corner) + free C_{p^k} covers of degree p^k.
- **Quinn $I(X)$:** local L-signature measuring resolution — substitutes for a missing Euclidean ball on BFMW ENRs.
- **Characters / τ:** covering-degree bookkeeping (OpenAI Fourier language; Dabrowski permutation modules).
- **Algebra:** integrality of Witt/L tails vs unbounded p-splitting of the unit.

Same obstruction class — once you see the germ + covers, the three formalisms stop looking unrelated.

AI-assisted / AI-generated with Cursor (Grok, Anthropic, and OpenAI models). Direction: Alan J. Coppola.